INCH-POUND

MIL-P-70762 (AR)
12 April 1990

MILITARY SPECIFICATION

PROPELLANT, COMPOSITE, CAST, FOR USE IN SIMULATOR, MISSILE: Sagger XM27

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements and quality assurance provisions for the manufacture and acceptance of composite cast propellant (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specification, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and Supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-192	- Ammonium Perchlorate, Technical
MIL-D-6998	- Dichloromethane, Technical
DOD-D-23443	- Di-(2-Ethylhexyl) Adipate, Technical
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specifications For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
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MIL-F-82655 - Ferric Oxide, Technical
MIL-A-85495 - Antioxidant, T-Butylphenol Type 2, 2'- Methylenebis

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1171 - Acceptance and Description Sheets (For Propellants and Explosives)
DOD-STD-2101 - Classification of Characteristics
MIL-STD-45662 - Calibration System Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Military Specifications and Standards, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified the issues shall be those in effect on the date of the solicitation.

SPECIFICATIONS

NAVAL SEA SYSTEMS COMMAND

WS 20700 - Polybutadiene, Liquid, Hydroxyl Terminated

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

12576572 - Simulator, Missile: Sagger XM27
12910159 - Dicyclohexylmethane 4, 4'- diisocyanate
12910160 - Cure Catalyst; Triphenyl Bismuth

(Copies of specifications, standards, other government documents, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by contracting officer.)

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2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issues of the nongovernment documents which is current on the date of the solicitation.

CHEMICAL PROPULSION INFORMATION AGENCY

CPIA, Publication No. 21 - The Solid Propellant Mechanical Behavior: Method for Determining Uniaxial Tensile Test at Constant Strain Rates

(Application for copies should be addressed to: Chemical Propulsion Information Agency, Applied Physics Laboratory, John Hopkins University, John Hopkins Road, Laurel, MD 20810).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 330 Average Particle Size of Refractory Metals and Compounds by Fisher Subsieve

ASTM D 792 Specific Gravity and Density of Plastics by Displacement

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Ingredients. All ingredients used in the manufacture of the propellant shall comply with the applicable specifications or drawings as follows:

Constituent Materials	Conforming to Specifications or Drawings
Hydroxyl terminated polybutadiene, (HTPB), Type II	WS 20700

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Constituent Materials	Conforming to Specifications or Drawings
2, 2'-Methylenebis-(4-methyl-6-tertiary-butylphenol), (DMMBP)	MIL-A-85495
Dicyclohexyl methane 4, 4'-diisocyanate, (H_{12}MDI)	12910159
Di-(2-ethylhexyl) adipate, (DOA)	DOD-D-23443
Triphenyl bismuth	12910160
Ferric oxide (Red), ($Fe_2O_3 . H_2O$)	MIL-F-82655
Ammonium perchlorate, (AP), Grade C, bimodal blend	MIL-A-192

3.2 Mixer composition requirements.

3.2.1 Mixer composition. The composition of the propellant being added to the mixer shall meet the requirements of Table I.

Table I. Composition of materials being added to the mixer

Material	Requirement (% by weight)	Composition Test Method
HTPB and DMMBP mix	see 4.5.1	4.5.3
H_{12}MDI	see 4.5.2	4.5.3
DOA	2.992 to 3.008	4.5.3
Triphenyl bismuth	0.029 to 0.031	4.5.3
Ferric oxide	1.995 to 2.005	4.5.3
Ammonium perchlorate (AP)	72.635 to 73.365	4.5.3

3.2.2 AP particle size. The AP used in the manufacture of this propellant shall meet the particle size requirements given in Table II. AP ratio may be varied within the limits of Table II to obtain the requirements of 3.3.3.

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Table II. AP particle size blend ratio

AP Particle Diameter	Blend ratio requirement (% by weight)	Applicable Test Method
MIL-A-192, Grade C Ground to an average diameter of 6 ± 2 microns	61.37 - 75.00	4.5.4
MIL-A-192, Grade C Class 4, 200 microns nominal size	25.00 - 38.63	4.5.4

3.3 Chemical, mechanical, and ballistic properties. The cured propellant samples used to determine the density, mechanical properties, strand burning rate, heat of explosion, and stability shall be stored at 77 ± 10 degrees F and 30 - 60% relative humidity in a condition where moisture absorption is minimized. Excursions from these conditions for periods up to 8 hours for handling, transportation and machining are permissible. The propellant samples used for this testing shall not exceed a stored life of 4 weeks between the end of cure and the start of analysis.

3.3.1 Density. The density of the cured propellant shall be 1.55 ± 0.03 g/cm^3 at 77 ± 3 degrees F when determined in accordance with 4.5.5.

3.3.2 Mechanical properties. The mechanical properties of the cured propellant shall be as follows when determined in accordance with 4.5.6.

a. Tensile Strength: 100 pounds per square inch (psi) minimum

b. Elongation: Not less than 8 percent at maximum tensile load

3.3.3 Strand burning rate. The burning rate (r_b), in inches per second, shall fall within the band specified by the following equation when determined in accordance with 4.5.7.

$$r_b = AP^{0.4} \text{ inches/second}$$

where $A = 0.030 \pm 0.006$ ((inches/second) $(\text{psia})^{-0.4}$)
P = average actual pressure (psia)

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3.3.4 Heat of explosion. The heat of explosion of the cured propellant shall be 890 ± 50 calories per gram (cal/g) when determined in accordance with 4.5.8. and 4.5.8.1.

3.3.5 Stability. A vaccuum stability test shall be done in accordance with 4.5.9 for informational purposes only.

3.4 Process controls. The contractor shall submit a Process Control Document to the Government specifying the process variables which are considered critical for the production of the propellant for this specification. The Process Control Document shall be submitted in accordance with 4.3.4.

3.5 Description sheets. The contractor shall prepare a sample description sheet and sample propellant acceptance sheet, modifying the format given in MIL-STD-1171, to contain the information needed to show that the manufactured propellant has met the requirements of this specification. The contractor shall submit the sample propellant description sheet and sample propellant acceptance sheet to the government technical agency (see 6.4) for approval before using these sheets.

A copy of the propellant description sheet and propellant acceptance sheet for each manufacture lot of propellant shall be provided to the technical agency (see 6.4).

3.6 Workmanship. The manufacturer shall use procedure and controls which assure that the propellant is free from any contamination or foreign material that would render the propellant unsuitable for the propellant's intended use (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless ~~disapproved by the~~ Government. The Goverment reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein, or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorized submission of known defective material, either indicated or actual, nor does it commit the Goverment to acceptance of defective material.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First Article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of five (5) blocks of propellant which have been vaccuum cast and cured in accordance with note two (2) of drawing 12576572. Each block shall be large enough for all first article tests of section 4.3.2, but shall not be larger than 700 cubic inches. One of the five samples shall be cast from the first mixer load and one from the last mixer load of the propellant lot. The three remaining samples shall be randomly taken from the propellant mix if the lot contains more than one mixer load. If the lot is one mixer load, then one sample shall be taken from the top, one from the bottom, and three from the middle. These samples shall be obtained from the first production lot which has been produced by the contractor using the same production process, procedures and equipment as will be used in fulfilling the contract. All material shall be [illegible] the same sources of supply as will be used in

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MIL-G-5

ator set shall be examined for the
se applicable ones from MIL-G-28554.

ified type used for preservation of
l C (5.1.1 and 5.1.2).
C not in accordance with the alternate

s not protected with wood cover as

ings, if so equipped not sealed with

4.9.2 Examination. Each packaged gener
following major defects in addition to tho

163. Anti-freeze of other than spec
cooling system for levels A an
164. Engine preservation for level
procedures specified (5.2.2).
165. Generator sets without housing
specified for level A (5.2.1).
166. Openings of generator set hous

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4.3.3 Rejection. See MIL-A-48078.

4.3.4 Process control. A Process Control Document shall be submitted to the Technical Agency (see 6.4). The document shall contain a description of the process, all materials used, process conditions/procedures, production/inspection equipment used, and all test methods used to ensure product quality. In addition, whenever there is a change (change in source of materials, process conditioning procedures, etc.) from previously reported process control data, the Technical Agency shall be notified of the new process control data and the reason for the change.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. A lot shall consist of one or more batches of propellant produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continous set of operation conditions. Each lot shall consist of that quantity of propellant that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. The lot shall comply with the provisions for submission of product as specified in MIL-STD-105. The criteria and procedure for the assignment of lot number shall be in accordance with MIL-STD-1168.

4.4.2 Examination. Unless otherwise specified in the Classification of Defects and test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. See MIL-A-48078.

4.4.3 Classification of characteristics. The classification of defects shall be as given in Table III.

PRECAUTION: This specification covers sampling and testing of chemical or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures shall be followed in handling and processing these materials.

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TABLE III. Classification of Defects

Test Examination	Paragraph	Defect Classifications
Mixer composition weights	3.2.1	Major
AP particle size	3.2.2	Major
Density	3.3.1	Major
Mechanical properties	3.3.2	Major
Strand burning rate	3.3.3	Major
Heat of explosion	3.3.4	Major
Stability	3.3.5	Major

4.4.4 Quality conformance sampling and testing. The quality conformance sample shall consist of three (3) blocks of propellant which have been vaccuum cast and cured in accordance with note two (2) of drawing 12576572. Each sample shall be large enough to do the testing required but not larger than 300 cubic inches. One of the three samples shall be cast from the first mixer load and one from the last mixer load of the propellant lot. The remaining sample shall be randomly taken from the propellant mix if a lot contains more than one mixer load. If the lot is one mixer load, then one sample shall be taken from the top, one from the bottom and one from the middle. Each of the five samples shall be subjected to all the tests of section three and section four of this specification. If any of the samples do not meet the requirements of section three, the lot shall be rejected.

4.4.5 Inspection equipment. For the performance of all tests and examinations specified in section 4, commercial inspection equipment should be employed. The contractor shall have available and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see 6.3). Unless otherwise specified, these requirements shall have the following minimum accuracies:

a. Time measurements: ± 0.05 percent

b. Force measurements: ± 1.0 percent

c. Pressure measurements: ± 1.0 percent

4.5 Methods of inspection (see 6.5).

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4.5.1 HTPB and DMMBP mixture determination (further clarification see 6.6). Use an established laboratory analysis method to determine the -NCO (isocyanate) and -OH (hydroxyl) equivalent weights for the lots of H_{12} MDI and HTPB to be used in the propellant mix.

Solve the following equation to determine the % HTPB and DMMBP mix requirement for section 3.2.1.

$$X = \frac{21.97}{(\text{cure ratio})\ ((\text{OH equivalents}/100\ \text{grams}) + 0.00588)} \over (1.01)\ (\text{NCO equivalents}/100\ \text{grams})$$

where: X = % HTPB and DMMBP mix
cure ratio = the cure ratio may be varied, within 0.90 to 1.10, so that physical properties are met (cure ratio is determined by the propellant manufacturer)

The % HTPB is then determined from the following equation.

$$X_1 = \frac{X}{1.01}$$

where: X_1 = % HTPB

X = % HTPB and DMMBP mixture

The % DMMBP is then determined from the following equation.

$$X = X_2 + X_1$$

where: X_2 = % DMMBP

X = % HTPB and DMMBP mixture

X_1 = % HTPB

4.5.2 H_{12} MDI determination. (see 6.6.). Determine the % H_{12} MDI requirement using the following equation.

$$Y = (21.97 - X)$$

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where: Y = % H_{12} MDI

X = % HTPB and DMMBP mix, solved for in section 4.5.1.

4.5.3 Composition weight check. Determine the weight of each of the materials being added to the mixer by multiplying the % by weight divided by 100 (see requirements of Table I) times the total amount of material that shall be added to the mixer. Triple weigh each container (all three weights must be exactly the same) with the material in it, before addition to the mixer, to the required decimal place (see Table I). Record these values. After addition of the material to the mixer, triple weight (all three weights must be the same) the container to the required decimal place. Subtract the before and after weighings to determine the amount of material added to the mixer. The weight shall meet the requirements of Table I, all weights shall be recorded on the propellant description sheet(s).

4.5.4 AP particle size and blend ratio determination. Randomly remove a two pound sample of AP from each lot to be used in the manufacture of the propellant. Pass the AP through the appropriate sieves in order to determine if the particle size requirements of Table II are met. Record the results on the propellant description sheet(s). Use only AP which meets the requirements of Table II in the manufacture of the propellant.

The AP blend shall be prepared by triple weighing (before and after blend preparation) all the containers and materials used in making the blend ratio. All weights shall be recorded on the propellant description sheets.

4.5.5 Density determination. The density of the propellant shall be determined in accordance with ASTM D 792. The test temperature shall be 77 ± 3 degrees F.

4.5.6 Mechanical properties test. Tensile strength and elongation at maximum tensile load shall be determined on Class B or Class C tensile test specimens in accordance with the procedures for preparation, storing, conditioning, and testing of tensile specimens given in section 4.3 of CPIA Publication No. 21, Class B - "Quality Control Testing" or Class C - "Development Screening Testing." These two methods shall give equivalent results for the material. The specimen shall be tested using a crosshead separation rate of 2 inches per miniute and at a test temperature of 77 ± 2 degrees F and 85 percent maximum relative humidity. An effective gauge length of 2.7 inches shall be used to calculate elongation.

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4.5.7 Strand burning rate. Strand burning rate on cured specimens shall be determined according to MIL-STD-286, Method T803.1. Strands shall be inhibited with polyester inhibitor (Koppers 6065-5 Polyester Resin catalyzed with 1-1/2 percent MEK Peroxide (60 percent MEK peroxide in dimethylphthalate) or equivalent). A minimum of three strands shall be tested at each of three pressures: 15 + 5, -0, 30 ± 5, and 45 ± 5 psia. Strands shall be burned at 77 degrees ± 3 degrees F. The starting pressure and the maximum pressure attained shall be recorded and averaged. The pressure gage for this test shall have an accuracy of 1 psia and the timer shall have an accuracy of 0.5 seconds.

4.5.8 Heat of explosion. This test shall be performed using an adiabatic calorimeter such as the Parr Instrument Company, Series 1200. A Parr No. 1101 or 1108 double value oxygen bomb, or equal, shall be used. The calorimeter shall be standardized with benzoic acid using oxygen according to the manufacture's instruction manual. The heat of explosion test shall be conducted in the same manner as the standardization with the following limitations and exceptions.

a. Water shall not be added to the bomb.

b. The bomb shall be purged three times with 15 to 20 atmospheres of helium or argon; then pressurize the bomb to 25 ± 2 atmospheres of helium or argon for testing.

NOTE: Impurities in the inert gas such as oxygen or water can cause systematic high results. Each new bottle should be tested for suitability for this test by comparison with gas of known purity.

4.5.8.1 Heat of explosion calculation.

$$\text{Heat of explosion, cal/gm} = \frac{(t_2 - t_1)W}{m}$$

where:

t_2 = temperature of calorimeter at end of test, degrees Celsius (C).

t_1 = temperature of calorimeter before ignition, degrees Celsius (C).

W = energy equivalent factor determined by standardization as above, calories per degrees C.

m = weight of sample, gram(s).

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4.5.9 Vaccuum stability test. Determine the vaccuum stability of the propellant at 90 degrees C using Method 403.1.2 of MIL-STD-286.

5. PACKAGING

(This section is not applicable to this specification.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This cast propellant is intended for use in the Missile Simulator, Sagger XM27, a training round.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval See MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAR-R (D), Picatinny Arsenal, NJ 07806-5000.

6.4 Technical agency. Submit all technical correspondence to Commander, US Army Armament Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAR-R (D), Picatinny Arsenal, NJ 07806-5000.

6.5 Equivalent test methods. The test methods given in this specification are official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, US Army Armament Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAR-R (D), Picatinny Arsenal, NJ 07806-5000. This description should include, but not be limited to the procedures used, the accuracy and precision of the method, test date to demonstrate the accuracy and precision and drawings of any special equipment required.

6.6 Composition calculation. The cure ratio may be varied, within 0.90 to 1.10 if needed, so that physical properties are met.

$$\text{Cure Ratio} = \frac{(\text{NCO equivalents/100 grams})(\%H_{12}\ \text{MDI})}{(\text{OH equivalents/100 grams})(\%\text{HTPB}) + 0.588)(\%\text{DMMBP})}$$

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where:

HTPB and DMMBP is 100 part HTPB to 1 part DMMBP mixture, and the sum of the percentages of H_{12}MDI, HTPB, and DMMBP is equal to the total percentage of binder in propellant composition (21.97%).

Therefore, the percentages of H_{12}MDI, HTPB, and DMMBP used in each particular mix are determined as follows:

a. The -NCO (isocyanate) and -OH (hydroxyl) equivalent weights are determined by laboratory analysis for the lots of H_{12}MDI and HTPB to be used in the mix.

b. The following three equations are solved for percent IDPI and percent HTPB & DMMBP mixture.

(1) $$\frac{(\text{NCO equivalents/100 grams})\ (\%H_{12}\text{MDI})}{(\text{OH equivalents/100 grams})\ (\%\text{HTPB}) + 0.588)\ (\%\text{DMMBP})}$$

(2) $$(\%\text{HTPB \& DMMBP mixture}) = (1.01)\ (\%\text{HTPB})$$

(3) $$(\%H_{12}\text{MDI}) + (\%\text{HTPB \& DMMBP mixture}) = 21.97\%$$

A sample calculation using a cure ratio of 1.026 is as follows:

H_{12}MDI sample analyzed is 0.759 - NCO equivalents/100 grams

HTPB sample analyzed is 0.081 - OH equivalents/100 grams

Let:

X = % HTPB & DMMBP mixture
X_1 = % HTPB
X_2 = % DMMBP
Y = % H_{12}MDI

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Then:

$X_1 + X_2 = X$; $X = 1.01X_1$; and $X + Y = 21.97$

and

$$\frac{(0.759)\ (21.97 - X)}{0.081\ (X/1.01) + 0.588\ (X - X/1.01)} = 1.026$$

or

$$X = \frac{21.97}{\frac{(1.026)\ (0.081 + 0.00588)}{(1.01)\ (0.759)} + 1.0} = 19.681$$

therefore

X_1 = 19.681/1.01 = 19.486%

X_2 = 19.681 - 19.486 = 0.195%

Y = (21.97 - X) = 2.289%

6.7 Solvent for triphenyl bismuth. Dichloromethane, in accordance with MIL-D-6998, may be used as a solvent for the triphenyl bismuth. This solvent shall be removed (evaporated) from the composition during mixing at low pressure.

6.8 Immersion liquid. N-hexane has been found to be a satisfactory immersion liquid.

6.9 Subject term (key word) listing.

Propellant
Rocket propellant
Sagger simulator propellant

Custodian
Army-AR

Preparing Activity:
Army-AR

(Project 1376-A360)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-70762 (AR)	2. DOCUMENT DATE (YYMMDD) 12 April 1990

3. DOCUMENT TITLE
PROPELLANT, COMPOSITE, CAST, FOR USE IN SIMULATOR, MISSILE: Sagger XM27

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army ARDEC Standardization & Specification Office	b. TELEPHONE (Include Area Code) (1) Commercial: (201) 724-6671 (2) AUTOVON: 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S/Bldg 6 Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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